

Searching Tables.

Creating Pre-filled Tables

A B C D E F G H I J K L

```
01 LetterTable.  
  02 TableValues.  
    03 FILLER PIC X(13)  
      VALUE "ABCDEFGHIJKLM".  
    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".
```

Creating Pre-filled Tables

A	B	C	D	E	F	G	H	I	J	K	L	
---	---	---	---	---	---	---	---	---	---	---	---	--

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02 FILLER REDEFINES TableValues.  
  03 Letter PIC X OCCURS 26 TIMES.
```

Searching a Table

A	B	C	D	E	F	G	H	I	J	K	L	
1	2	3	4	5	6	7	8	9	10	11	12	

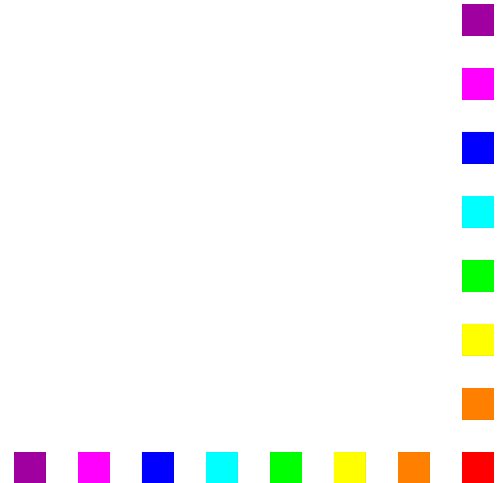
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    03 Letter PIC X OCCURS 26 TIMES.
```

```
PERFORM VARYING Idx FROM 1 BY 1 UNTIL  
  LetterIn EQUAL TO Letter(Idx)  
END-PERFORM.  
DISPLAY LetterIn, "is in position ", Idx.
```

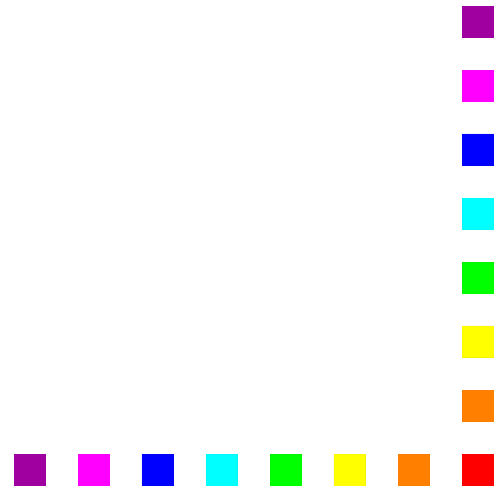
Search Syntax

$$\begin{aligned} & \underline{\text{SEARCH}} \quad \text{TableName} \left[\underline{\text{VARYING}} \left\{ \begin{array}{l} \text{Identifier} \\ \text{IndexName} \end{array} \right\} \right] \\ & \quad [\text{AT } \underline{\text{END}} \text{ StatementBlock}] \\ & \quad \left\{ \underline{\text{WHEN}} \text{ Condition} \left\{ \begin{array}{l} \text{Statementblock} \\ \underline{\text{NEXT SENTENCE}} \end{array} \right\} \right\} \dots \\ & \underline{\text{END-SEARCH}} \end{aligned}$$

OCCURS TableSize TIMES
[INDEXED BY {IndexName}...]



SET Syntax



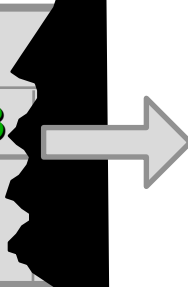
Searching a Table

A	B	C	D	E	F	G	H	I	J	K	L	
1	2	3	4	5	6	7	8	9	10	11	12	

```
01 LetterTable.  
  02 TableValues.  
    03 FILLER PIC X(13)  
      VALUE "ABCDEFGHIJKLM".  
    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".  
  02 FILLER REDEFINES TableValues.  
    03 Letter PIC X OCCURS 26 TIMES  
      INDEXED BY LetterIdx.  
SET LetterIdx TO 1.  
SEARCH Letter  
  AT END DISPLAY "Letter not found!"  
  WHEN Letter(LetterIdx) = LetterIn  
    DISPLAY LetterIn, "is in position ", Idx  
END-SEARCH.
```

Searching a Two Dimension Table.

1								2								3										
1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	1	2	3



```
01 TimeTable.
  02 Day OCCURS 5 TIMES INDEXED BY DayIdx.
    03 Hours OCCURS 8 TIMES INDEXED BY HourIdx.
      04 Item          PIC X(10).
      04 Location      PIC X(10).

SET DayIdx TO 0.
PERFORM UNTIL MeetingFound OR DayIdx > 5
  SET DayIdx UP BY 1
  SET HourIdx TO 1
  SEARCH Hours WHEN MeetingType = Item(DayIdx, HourIdx)
    SET MeetingFound TO TRUE
    DISPLAY MeetingType " on " DayIdx " at " HourIdx
  END-SEARCH
END-PERFORM.
```


Search All Syntax.

SEARCH ALL TableName [AT END StatementBlock]

$$\begin{array}{l}
 \text{WHEN} \left\{ \begin{array}{l} \text{ElementIdentifier} \left\{ \begin{array}{l} \text{IS } \underline{\text{EQUAL TO}} \\ \text{IS } = \end{array} \right\} \left\{ \begin{array}{l} \text{Identifier} \\ \text{Literal} \\ \text{ArithExpression} \end{array} \right\} \\ \text{ConditionName} \end{array} \right. \\
 \left[\begin{array}{l} \text{AND} \left\{ \begin{array}{l} \text{ElementIdentifier} \left\{ \begin{array}{l} \text{IS } \underline{\text{EQUAL TO}} \\ \text{IS } = \end{array} \right\} \left\{ \begin{array}{l} \text{Identifier} \\ \text{Literal} \\ \text{ArithExpression} \end{array} \right\} \\ \text{ConditionName} \end{array} \right\} \end{array} \right] \\
 \left\{ \begin{array}{l} \text{StatementBlock} \\ \text{NEXT SENTENCE} \end{array} \right\}
 \end{array}$$

END – SEARCH

OCCURS TableSize TIMES

[{ ASCENDING } KEY IS {ElementIdentifier} ...] ...
 [INDEXED BY {IndexName} ...]

Using the Search All

A	B	C	D	E	F	G	H	I	J	K	L	
1	2	3	4	5	6	7	8	9	10	11	12	

```
01 LetterTable.  
  02 TableValues.  
    03 FILLER PIC X(13)  
      VALUE "ABCDEFGHIJKLM".  
    03 FILLER PIC X(13)  
      VALUE "NOPQRSTUVWXYZ".  
  02 FILLER REDEFINES TableValues.  
    03 Letter PIC X OCCURS 26 TIMES  
      ASCENDING KEY IS Letter  
      INDEXED BY LetterIdx.  
  
SEARCH ALL Letter  
  WHEN Letter(LetterIdx) = LetterIn  
    DISPLAY LetterIn, "is in position ", Idx  
END-SEARCH.
```

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Lower

1

Upper

26

Middle

13

=

Letter(Middle)

M

ALGORITHM.

$\text{Middle} = (\text{Lower} + \text{Upper}) / 2$

CASE TRUE

WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1

WHEN Letter(Middle) > "Q" THEN Upper = Middle - 1

WHEN Letter(Middle) = "Q" THEN SET ItemFound TO TRUE

WHEN Lower > Upper THEN ItemNotInTable TO TRUE

How the Search All works.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

14

Upper

26

Middle

13

=

Letter(Mi

ddle)

M

ALGORITHM.

Middle = (Lower + Upper) / 2

CASE TRUE

WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

14

Upper

26

Middle

20

=

Letter(Mi

ddle)

T

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CASE TRUE

WHEN Letter(Middle) < "Q" THEN Lower = Middle + 1

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Lower

14

Upper

19

Middle

20

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1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Lower

14

Upper

19

Middle

16

=

Letter(Mi

ddle)

P

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18

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Search All Example.

```
01 StateTable.  
  02 StateValues.  
    03 FILLER PIC X(20) VALUE ??????????????  
        Post Codes and Names  
    03 FILLER PIC X(20) VALUE ??????????????  
  
  02 FILLER REDEFINES StateValues.  
    03 States OCCURS 50 TIMES  
        ASCENDING KEY IS StateName  
        INDEXED BY StateIdx.  
        04 PostCode      PIC X(6).  
        04 StateName     PIC X(14).  
  
SEARCH ALL States  
  AT END DISPLAY "State not found"  
  WHEN StateName(StateIdx) = InputName  
    MOVE PostCode(StateIdx) TO PrintPostCode  
END-SEARCH.
```